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GRADE SERIES

GRADE Guidance 44: strategies to enhance the utilization of randomized and non-randomized studies in evidence syntheses of health interventions

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Abstract

Background and Objectives: Ideally, guideline developers and health technology assessment authors base intervention decisions on randomized controlled trials (RCTs). However, relying solely on RCTs is uncommon, especially for public health interventions and harms assessment. In these situations, nonrandomized studies of interventions (NRSIs) can provide valuable information. This article presents Grading of Recommendations Assessment, Development, and Evaluation (GRADE) guidance for integrating bodies of evidence RCT and NRSI in evidence syntheses of health interventions.

Methods: Following standard GRADE methods, we developed this guidance through iterative discussions and examples with experts from the GRADE NRSI project group in multiple dedicated meetings. We presented findings of the group discussions for feedback at GRADE Working Group meetings in September 2023 and May 2024.

Results: The resulting GRADE guidance outlines a structured approach: (1) assessing the certainty of evidence (CoE) after defining the number of decision thresholds and the target of the certainty rating; (2) evaluating congruency of effect estimates between RCTs and NRSIs; (3) identifying which GRADE domains are affected by certainty ratings to inform complementariness between RCTs and NRSIs and the overall CoE; and (4) deciding whether and how to use one or both types of studies.

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Background for this work has been obtained through Surveys and discussions within the GRADE NRSI/RCTs integration working group. Slides and case studies have been presented throughout different GRADE meetings in Washington, D.C., U.S. (2016), Seoul, South Korea (2016), Rome, Italy (2017), Hamilton, Canada (2017), Krakow, Poland (2022), London, UK (2023), Glasgow, UK (2023), Miami, USA (2024), and online meetings in between.

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Conclusion: This GRADE guidance offers a structured and practical approach for integrating or not integrating RCTs and NRSIs in evidence syntheses. By addressing the interplay between affected GRADE domains and assessing the congruency of effects, it helps GRADE users determine when and how NRSIs can meaningfully complement or replace RCT evidence to inform certainty ratings and decision-making. © 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Keywords: GRADE; Certainty of evidence; Risk of bias; Evidence synthesis; Nonrandomized studies; Randomized controlled trials

1. Introduction

High-certainty evidence from randomized controlled trials (RCTs) allows for confident conclusions from evidence syntheses assessing health interventions. Well-conducted RCTs, ie, those with an adequate design, number of representative participants, follow-up, measure of outcomes, etc, will yield effect estimates that are less susceptible to bias and other sources of uncertainty procuring better opportunities to answer questions of causal effects of interventions [1]. These instances are desirable but remain relatively rare and may explain why a sizable proportion of the research dedicated to health interventions relies on observational evidence [2−4].

Evidence syntheses of health interventions, whether these are standalone evidence synthesis or used to inform health technology assessments (HTAs) or health care guidelines, include evidence comprised of RCTs, nonrandomized studies of interventions (NRSIs), or a combination of both.

Previous Grading of Recommendations Assessment, Development, and Evaluation (GRADE) publications have examined the use of RCTs and NRSI in evidence synthesis, including the rating of the certainty of evidence (CoE) in NRSIs and the use of specific risk of bias (RoB) tools [5] authors' practices and preferences, [6] and case studies exploring differences between RCTs and NRSI [7]. This body of work informed earlier GRADE guidance on when NRSI should be considered alongside RCTs in syntheses addressing interventions [8,9].

GRADE users who consider utilizing both RCTs and NRSIs to inform a given question require guidance on how to effectively integrate evidence from these 2 distinct study designs. This official guidance aims to support such integration by providing structured principles to evaluate when and how combining RCT and NRSI evidence is appropriate within the GRADE approach.

2. Objectives

To provide researchers and decision-makers with a practical framework for integrating NRSIs with RCTs in health intervention assessments, aiming for the highest possible CoE.

3. Materials and methods

3.1. Overview

This work, part of the GRADE project group on RoB and NRSI, involved identifying examples, iterative discussions, and presentations at GRADE Working Group meetings (2018−2023) following GRADE's guidance development approach [10].

3.2. Operationalization of guidance

We examined methods for evidence rating, result presentation, and the GRADE domains most affected by integrating RCTs and NRSIs. Building on prior work on author attitudes toward pooling these studies, we conducted in-depth discussions and multiple virtual meetings, including 2 full-day sessions.

We identified key examples to guide discussions on when to synthesize RCT and NRSI evidence. We defined scenarios for using NRSIs as replacement, complementary, or sequential evidence in certainty ratings. Recognizing the importance of effect estimate similarity, we focused on defining congruency between RCTs and NRSIs and its impact on deciding which evidence to use and how to rate the certainty [8,9].

We summarized the results for the relevant GRADE project group and presented the draft and final guidance to the full GRADE Working Group twice in September 2022 (40−60 attendees) and for approval in May 2024 (≈80 attendees). The GRADE Guidance Group formally approved the approach on May 2, 2024.

4. Guidance

4.1. Overview

In prior work, we have explored how GRADE assessments should be applied to NRSI [5], what strategies are used to assess GRADE domains in evidence syntheses that combine RCTs and NRSIs, and how these assessments influence the overall certainty of evidence (CoE) [7,11]. Building on those foundations and through extensive iterative discussions, we recognized that developing specific guidance for every possible combination of CoE levels and reasons for rating down — across domains such as study limitations (RoB),

What is new?

Key findings

- Provides official GRADE guidance for integrating RCTs and NRSIs in evidence syntheses using a structured, principle-based framework.
- Defines explicit processes to assess certainty of evidence and congruency of effect estimates across study designs.

What this adds to what is known?

- Extends previous GRADE publications by offering operational steps and decision flows for when NRSIs can complement, replace, or follow RCT evidence.
- Clarifies how domain-level judgments (risk of bias, indirectness, imprecision, etc.) interact when integrating evidence from different study designs.

What is the implication and what should change now?

- Gives guideline developers, HTA agencies, and evidence synthesis teams clear criteria for when integration is appropriate and when it should be avoided.
- Supports transparent and consistent judgments in contexts where evidence comes from mixed study designs.
- Apply the four-step framework to ensure that integrating RCTs and NRSIs is context-driven, methodologically sound, and clearly reported.
- Use the guidance to harmonize methods across decision-making settings and improve trust in recommendations informed by mixed evidence.

imprecision, inconsistency, indirectness, and dissemination bias — would be unfeasible. With 139 potential scenarios for RCTs and 140 for NRSIs, this yields 19,460 possible combinations. This complexity highlights the need for a structured, principle-based approach.

In the following sections, we outline 4 sequential steps to guide GRADE users when integrating RCTs and NRSIs in an evidence synthesis. The process begins with assessing the certainty of each body of evidence (Step 1), accompanied by guiding principles for evaluating relevant GRADE domains in both study designs. Step 2 focuses on determining the congruency of effects between RCTs and NRSIs, while Step 3 describes the decision-making process for comparing the two bodies of evidence across different scenarios, using the CoE from RCTs as the initial anchor point. Finally, Step 4 provides guidance on how to use and present RCT and NRSI findings within GRADE tables. The role of sensitivity analyses in supporting these assessments is also briefly discussed.

4.2. Step 1: assessing the CoE of both RCTs and NRSI

This guidance applies once GRADE users have determined that a body of NRSI evidence is relevant to the systematic review question and merits full consideration for integration alongside RCTs. This judgment follows the previous GRADE guidance [8], which helps users decide whether both study types address the same patient, intervention, comparison, outcome (PICO) question and are appropriate to consider together within an evidence synthesis. Building on that foundation, the present guidance suggests authors must assess the CoE for both RCTs and NRSIs individually — as high, moderate, low, or very low, according to current GRADE guidance outline in GRADE's official resource, the GRADE Book [12] — and then proceed to evaluate their combined value.

Before advancing in the evaluation of both bodies of evidence and making integration decisions, users should be familiar with the guiding principles developed through our iterative discussions. These principles, specific to each GRADE domain, underpin later steps in this process and inform final judgments about the overall CoE.

4.2.1. Guiding principles for evaluating individual GRADE domains in RCTs and NRSI

The principles here outlined should be applied together with the assessment of effect congruency (Step 2) and the decision scenarios anchored in the CoE of RCTs (Step 3). Collectively, these elements provide a coherent framework for determining when and how to integrate evidence from both study designs.

Table 1 presents a concise summary of the guiding principles for each GRADE domain and indicates when pooling may be appropriate. A detailed description of these principles — including rationale, illustrative examples, and extended explanations for each domain — is provided in the Supplementary Material 1 (section 1) to support reader understanding.

We use the term *mitigation* to describe circumstances in which a downgraded domain in one body of evidence can be reconsidered based on judgments from the other. For example, if RCTs are rated down one level for serious indirectness but NRSIs — without indirectness — show congruent results, the direct evidence from NRSIs may mitigate the indirectness in RCTs, restoring their rating to "not serious."

Table 1. Guiding principles for evaluating individual GRADE domains when integrating RCT and NRSI evidence

GRADE domain	Key principle	When pooling may be appropriate	Notes
Study limitations (risk of bias)	RoB in one body of evidence cannot be mitigated or "corrected" by lower RoB in the other.	Pool only when RoB levels are similar and imprecision is the only remaining concern.	Evaluate direction of bias; perform sensitivity analyses by study design.
Inconsistency	Inconsistency cannot be mitigated by consistency in the other body of evidence.	Do not pool if either body shows inconsistency.	NRSIs may help explain inconsistency in RCTs if plausible effect modifiers are identified.
Indirectness	A body of evidence without indirectness may help mitigate indirectness in the other, depending on other GRADE domains.	Do not pool if either body shows indirectness.	Examine differences in population, intervention, or baseline risk before judging congruency.
Imprecision	Imprecision in one body of evidence cannot be mitigated or "corrected" by no imprecision in the other.	Pool only to explore overall imprecision, provided there is no indirectness, inconsistency, or very serious RoB in either body of evidence.	Use absolute effects and decision thresholds to interpret overlap of CIs.
Dissemination bias	Publication bias in one body of evidence should not be mitigated by its absence in the other.	Considering both together (eg, funnel plots) may help explore the overall picture across bodies of evidence.	Assess each body independently; real-world examples are limited.

GRADE, Grading of Recommendations Assessment, Development, and Evaluation; NRSI, nonrandomized study of interventions; RCT, randomized controlled trial; RoB, risk of bias.

This table summarizes the guiding principles developed through iterative GRADE discussions. Each principle informs how individual GRADE domains should be applied when integrating RCT and NRSI evidence. Pooling should only be considered when justified and when other domain concerns are minimal.

4.3. Step 2: assessing congruency of effect estimates from RCTs and NRSI

Using NRSIs in the evidence synthesis depends on their value as replacement of RCTs, which is typically limited to cases where NRSI evidence has higher certainty. More commonly, however, is the case where NRSIs serve as complementary or sequential evidence [8,9], requiring an assessment of the degree to which effect estimates from RCTs and NRSIs align. This is a key consideration when integrating RCTs and NRSIs — essentially, whether they would support the same answer to the same PICO question. We refer to this alignment as *effect congruency*, and its absence as *incongruency*. Recent studies have applied various methods to assess congruency [4,13−15], ranging from comparisons of point estimates and CIs [16], classifying effects as statistically significant or not [17], conducting interaction tests within subgroup analyses [18], and comparing relative effects through ratios of odds ratios, risk ratios, or risk differences [13,14,19]. Some studies combine multiple approaches to their assessments [15]. Findings from these studies suggest that effect differences between RCTs and NRSIs are less common than initially assumed, and that when discrepancies arise, they may be influenced by factors beyond study design [13,14,19].

Statistical tests for detecting differences between RCT and NRSI estimates may have limited power [20] and do not account for effect magnitudes relative to important benefit or harm thresholds. To integrate uncertainty, magnitude of effects, and CoE, we defined congruency based on absolute effect thresholds and CIs, as informed by recent GRADE guidance and real-world case studies. These developments enable a straightforward determination of thresholds [21−23].

The GRADE Evidence to Decision (EtD) framework categorizes effects as trivial or no effect, small, moderate, or large, with corresponding certainty ranges. The Supplementary Material 1 (Figure S1 in section 2) illustrates how thresholds and magnitudes of effects relate in a typical forest plot from an evidence synthesis with meta-analysis.

We use multiple thresholds (6) as in the EtD to highlight how incongruency between a small and a large effect can be as impactful for decision-making as a difference between a trivial and a small (or important) effect.

Congruency is present when both effect absolute estimates and their CIs from RCTs and NRSIs fall within the same effect magnitude range, as seen in Figure 1A-D. Incongruency occurs when absolute effect estimates and CIs are within different effect magnitude ranges (Figure 1E-H). This approach is also adaptable when fewer thresholds are considered, such as when focusing solely on small effects (determine by the minimally important difference). The implication of using fewer thresholds, however, is that effects may erroneously be considered congruent when they are not (Figure 1G).

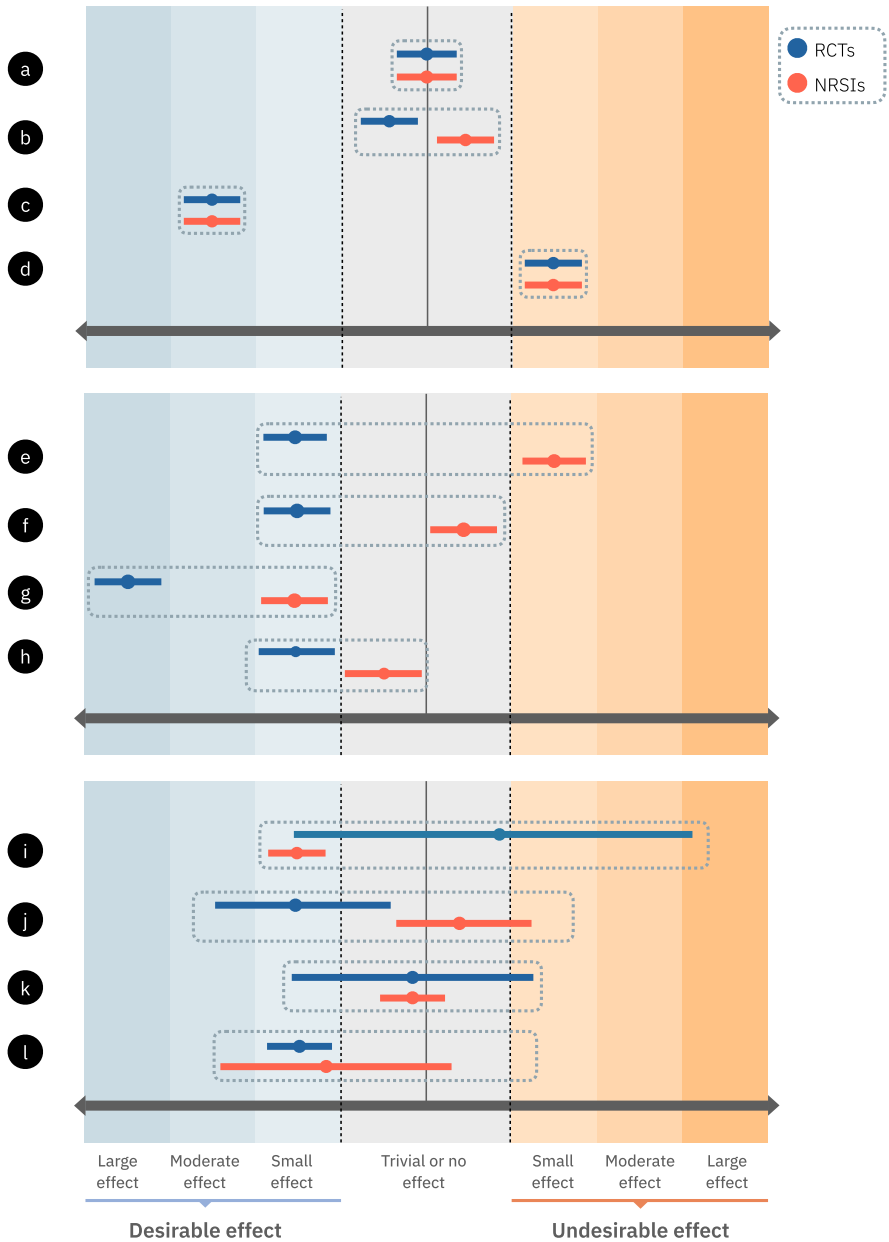

Figure 1. Different scenarios of congruency/incongruency between effect estimates of bodies of evidence of RCTs and NRSIs Illustration of 3 groups of scenarios using classic forest plots, with 6 thresholds that define the zones for magnitude of effect (as described in Fig. S1), to reflect congruency or incongruency in effect estimates between RCTs and NRSIs (see main text for details). Scenarios A, B, C, and D are considered congruent. Scenarios E, F, G, and h are incongruent, while scenarios I, J, K, and l represent situations of probable incongruency (I and J) or probable congruency (K and L) due to imprecision (wide CIs). GRADE users may refer to the flowchart in Figure 2 to help distinguish between these classifications. In scenario g, RCTs and NRSI are incongruent. However, if GRADE users consider only one threshold, ie, the threshold of small but important effect (minimally important difference), these will be classified as congruent, but it bears the risk of combining study designs despite wide variation in the study results (small vs large effects). Note that for congruency assessment both RCTs and NRSI require utilization of the same baseline risk to avoid incongruency based on baseline risk differences. GRADE, Grading of Recommendations Assessment, Development, and Evaluation, NRSI, nonrandomized studies of interventions; RCT, randomized controlled trial.

When effect estimates from RCTs or NRSI are affected by imprecision (ie, wide CI's) establishing congruency becomes more challenging. In such cases, we classify the findings as *probable congruency* or *probable incongruency*. For example, in Figure 1I the effect estimate from RCTs is confined to the target certainty range of a small effect, while the CI from NRSIs spans the ranges of important benefit, trivial effect, and important harm. Due to the imprecision of the NRSIs estimate, it is unclear whether the 2 bodies of evidence are truly in agreement.

To support users in making these determinations, we provide a decision-making flowchart, depicted in Figure 2.

When assessing congruency, authors must use absolute effect estimates — not relative effects. Absolute effects are calculated from pooled relative effects and the control arm's baseline risk. However, baseline risks in RCTs may not always be applicable, as they can differ from those in NRSIs due to more restrictive selection criteria. For instance, RCTs may show a lower baseline risk for outcomes like death, while NRSIs reflect those of higher risk populations. In such cases, authors should derive more applicable baseline risks from large, representative observational studies, or their own syntheses, and apply the relative estimates from the RCTs and NRSI to these baseline risks to evaluate congruency [24].

Establishing congruency between RCTs and NRSIs supports their integration and informs how GRADE domains may interact across both bodies of evidence. In contrast, incongruent findings indicate that the 2 sources should not be combined in the decision-making process. Identifying incongruency early helps users avoid unnecessary complexity and instead focus on the body of evidence that offers the higher CoE for the specific PICO.

In the rare instance where RCTs and NRSIs yield incongruent effect estimates but are rated at the same level of certainty, users must apply nuanced judgment within the level of certainty to determine which body of evidence offers more confidence for the specific decision or context.

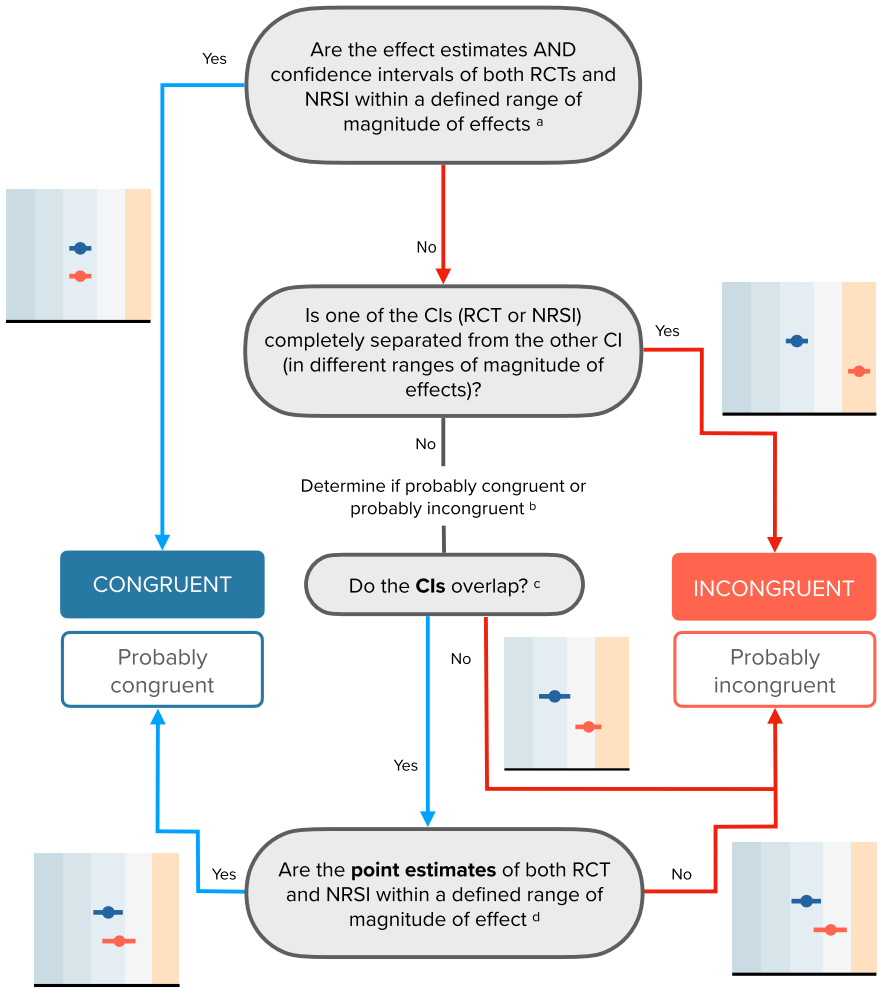

Figure 2. Flowchart for decisions about determining if effects from RCTs and NRSIs are congruent a. If effect estimates and CI's are completely within a range of a magnitude of effect (delimited by thresholds), we can be confident of its congruency. On the other hand, when RCTs and NRSI are completely separated by a threshold we can ascertain that they are incongruent. We use six thresholds to distinguish between trivial, small, moderate, and large effects (for benefits and harms). This simplified approach is also applicable at subsequent decision nodes in the flowchart. b. Imprecision in one or both CIs from RCTs and NRSIs introduces uncertainty when determining effect congruency. In such cases, authors must judge whether the findings are *probably congruent* or *probably incongruent*, as defined in the next steps of this flowchart. c. In most scenarios at this step, the CIs will overlap, and the answer to this question will be "yes." However, if a situation arises similar to Figure 1, example j, where overlap is minimal or absent, the answer would be "no." d. When CIs overlap but span into different magnitude-of-effect zones, the congruency of the effect estimates remains uncertain. In this case, the point estimates can help inform whether the scenario is better classified as *probably congruent* or *probably incongruent*, although uncertainty will remain. NRSI, nonrandomized studies of interventions; RCT, randomized controlled trial.

4.4. Step 3: comparing the CoE of RCTs and NRSIs

This guidance outlines key scenarios anchored in the CoE from RCTs, and the inclusion of NRSIs should be considered in each scenario, depending on their respective CoE. These are summarized in Table 2 and outlines how NRSI may complement, replace, or better remain as a separate body of evidence. Detailed explanations for each scenario are provided in the Supplementary Material 1 (Section 3).

For each scenario, users should also apply the guiding principles previously detailed in Section 4.2.1 and Table 1 to evaluate how NRSIs may contribute to the overall evidence synthesis and certainty.

Note: Exploring the use of both bodies of evidence with sensitivity analyses

Sensitivity analyses based on study design are well-established methods for testing the robustness of findings [25,26]. They involve comparing results when including or excluding specific study types (eg, RCTs vs NRSIs) or studies categorized by high vs low RoB, a commonly recommended and used strategy [27]. These approaches are particularly relevant when evaluating the potential limitations of NRSIs when integrating them with RCTs.

Sensitivity or subgroup analyses can also explore differences in other GRADE domains, such as indirectness or inconsistency. To ensure transparency and methodological rigor, all sensitivity analyses should be prespecified in the review protocol and conducted with input from both methodological and content experts.

4.5. Step 4: decide whether and how to use one or both bodies of evidence

The final step involves determining how to incorporate RCT and NRSI evidence into the overall synthesis and how to present it transparently in GRADE tables. This decision builds on the preceding steps — assessing CoE, evaluating effect congruency, and comparing both bodies of evidence applying domain-specific guiding principles throughout — and reflects the relative contribution of each body of evidence to the overall certainty rating. Depending on their final judgments, users may select among the following 3 presentation strategies:

1. Separate presentation: List RCTs and NRSIs separately in the GRADE tables without a pooled estimate, using 2

Table 2. Scenarios and suggested actions for integrating randomized and nonrandomized bodies of evidence based on their respective certainty

Certainty of evidence in RCTs	Certainty of evidence in NRSIs			
	High certainty	Moderate certainty	Low certainty	Very low certainty
RCTs: high certainty	Use RCTs and NRSIs together. Pooling into a single effect estimate is appropriate.	Rely on RCTs as the primary evidence. NRSIs may be considered as supplementary if they offer additional context or reinforce findings.	Use RCTs alone. In selected cases, NRSIs may be used as complementary evidence.	Use only RCTs. NRSIs are unlikely to contribute meaningfully and should generally not be considered.
RCTs: moderate certainty	Consider using NRSIs as a replacement for RCTs. Alternatively, consider integrating both, if justified, and assess how domains interact.	Consider presenting both. NRSIs may complement RCTs and jointly inform the certainty rating. Pooling may be appropriate depending on the GRADE domains affected.	Use RCTs as the primary evidence. NRSIs may be considered (eg, if they provide direct evidence for RCTs rated down for indirectness). Do not pool unless clearly justified.	Use only RCTs. NRSIs are unlikely to improve certainty and should not be used to mitigate limitations in RCTs. Presentation of NRSIs separately may be considered.
RCTs: low certainty	Rely on NRSIs as primary evidence. RCTs may be used to complement or support judgments depending on the affected GRADE domains. Pooling may be appropriate in some cases.	Rely on NRSIs as primary evidence. RCTs may be considered together as complementary. Pooling may be appropriate if justified and all domains are carefully assessed.	Consider presenting both. Integration ± pooling may raise certainty only if one body helps mitigate the other. Use caution when interpreting pooled estimates.	Rely on RCTs. Avoid pooling. Use NRSIs only if they contribute unique, context-specific information. In most cases, both bodies of evidence are insufficient for decision-making.
RCTs: very low certainty	Use NRSIs as the primary evidence. RCTs contribute little given their very low CoE. Do not pool unless clearly justified.	Rely on NRSIs as the leading source of evidence. RCTs may offer limited value. Do not pool unless clearly justified and domain interactions are understood.	Rely primarily on NRSIs. Note that their CoE remains limited. RCTs can be presented alongside for transparency. Avoid pooling. Interpret with caution.	Neither body of evidence provides sufficient certainty. Both can be presented, but base conclusions with transparency and acknowledge limitations. Avoid pooling.

CoE, certainty of evidence; NRSIs, nonrandomized studies of interventions; RCTs, randomized controlled trials.

The table illustrates overall guidance on how GRADE users may approach integration of RCT and NRSI evidence across varying levels of certainty. Decisions will depend on the interaction of affected individual GRADE domains as described in the text.

rows. This format applies whether results are presented as meta-analyses, narrative syntheses, or both.

2. Separate rows with a pooled estimate: When a pooled estimate is appropriate, include a third row showing combined RCT + NRSI data while retaining the 2 bodies of evidence as distinct subgroups.
3. Single-row presentation: Use one pooled estimate (without subgroups) or include only one body of evidence while describing the other in footnotes.

5. Decisions — putting all together

Figure 3 provides a summary of the full guidance here presented. The general approach for integrating RCTs and NRSIs to inform a review question involves the acknowledgment that NRSIs are relevant to the review question and warrant full consideration for integration alongside RCTs. By examining the guiding principles, GRADE users can evaluate how the affected domains interact across both bodies of evidence, and whether one helps reducing uncertainty in the other.

We emphasize the need to apply the guiding principles (Section 4.2) across all GRADE domains affected and consider sensitivity or subgroup analyses. Selecting the best presentation requires content and methodological expertise to understand interactions between GRADE domains and their impact on evidence synthesis conclusions. These decisions ultimately shape recommendations in guidelines, HTA panels, and other decision-making settings. In the Supplementary Materials 2 and 3 we present 2 worked examples applying this guidance of review questions using both RCTs and NRSIs. Finally, an infographic putting all guidance and explanations together can be found in the Supplementary Materials.

6. Discussion

We present comprehensive GRADE guidance for integrating NRSIs with RCTs as replacement, complementary, or sequential evidence. Despite its inherent complexity, the approach remains pragmatic, helping GRADE users navigate over 19,000 possible certainty rating scenarios. The process centers on assessing if effects between the 2 bodies

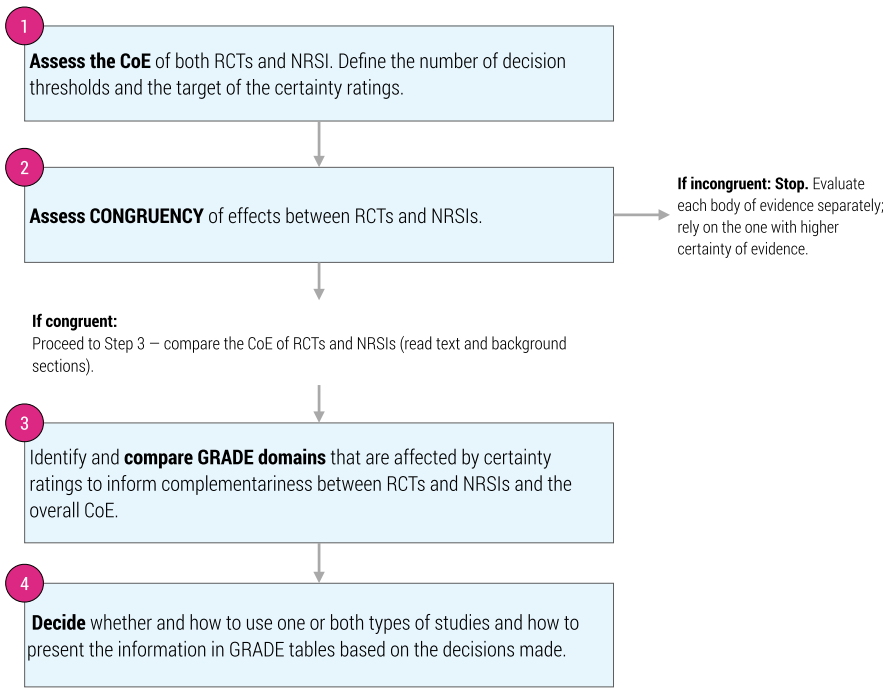

Figure 3. Overview of the four-step process for integrating RCT and NRSI evidence using the GRADE approach The figure summarizes the sequential process for integrating randomized controlled trials (RCTs) and nonrandomized studies of interventions (NRSIs) within an evidence synthesis. Infographic Footnotes (a) Consistent with prior guidance from the GRADE Working Group. b, These principles (recommendations) are essential for understanding the complete process and underlying rationale. See the main text and Supplementary Material for detailed explanations and illustrative examples. c, The steps of assessing congruency and CoE may be performed simultaneously. d, In systematic reviews, authors may still present both bodies of evidence separately to maintain transparency; however, only one should inform the conclusions —specifically, the body with the higher CoE, as it provides more trustworthy information for the PICO being evaluated. e, Scenarios in which NRSI evidence is rated as high certainty are exceedingly rare, as fully addressing confounding and other biases in nonrandomized designs is seldom feasible. f, When RCTs are rated as high certainty, additional information from NRSIs is generally unnecessary, since RCT estimates are considered sufficient and reliable for the specific PICO. g, Choose one of the three available GRADE table formats to present evidence profiles and/or summary of findings tables. Add footnotes as appropriate. These options also apply when a narrative synthesis is used instead of a pooled (meta-analytic) estimate. CoE, certainty of evidence; GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; HTA, health technology assessment; NRSI, nonrandomized study of interventions; PICO, population, intervention, comparison, outcome; RCT, randomized controlled trial; RoB, risk of bias.

of evidence are congruent before evaluating individual GRADE domains across RCTs and NRSIs.

6.1. Strengths and limitations

This official GRADE guidance reflects contributions from experts in GRADE, Cochrane, guidelines, and HTA communities, ensuring both methodological rigor and practical applicability. Its iterative derivation allowed for refinement through nuanced, real-world scenarios, making it adaptable to common challenges in evidence synthesis.

Some limitations remain, particularly when RCTs and NRSIs yield incongruent effect estimates yet are rated at the same level of certainty. In such (rare) cases, determining which body of evidence to rely on depends on nuanced assessments within a level of certainty using judgment (eg, if RoB is more important than imprecision) and contextual interpretation. Additionally, the complexity of all possible combinations of certainty ratings and affected domains highlights the impracticality of laying out every scenario individually in this article. We have adopted a pragmatic, principle-based approach to support reviewers now, while also acknowledging that further guidance, decision aids, and large language models (artificial intelligence)-assisted tools are in development to support more consistent and scalable applications.

6.2. Implications for decision-makers and guideline developers

This GRADE guidance provides a structured approach to integrate evidence from RCTs and NRSIs in evidence syntheses. The guidance applies broadly to various NRSI designs — from large cohorts to other types often inappropriately referred to as real-world evidence — especially useful when RCTs alone are insufficient. By helping users determine when integration is appropriate and when separate consideration is warranted, it promotes more transparent, consistent, and context-relevant decision-making. Future guidance will expand on key aspects that enhance confidence in NRSIs, such as directed acyclic graphs, statistical adjustment methods, assessment of bias direction, and specific NRSI designs. We also plan to address more advanced strategies for summarizing and presenting integrated evidence in GRADE tables and EtD frameworks.

7. Conclusions

We developed official GRADE guidance for authors incorporating NRSIs alongside RCTs in syntheses of health interventions. This work offers a practical approach to assess congruency, navigate interactions across GRADE domains, and support transparent decision-making when combining evidence from different study designs or deciding to focus on one design. The guidance is expected to improve consistency in how NRSIs are considered, particularly in areas where RCTs alone are insufficient and will be added to the GRADE Book as GRADE's final official resource.

CRediT authorship contribution statement

Carlos A. Cuello-Garcia: Writing − review & editing, Writing − original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rebecca L. Morgan:** Writing − review & editing, Conceptualization. **Nancy Santesso:** Writing − review & editing, Conceptualization. **Pablo Alonso-Coello:** Writing − review & editing, Conceptualization. **Romina Brignardello-Petersen:** Writing − review & editing, Conceptualization. **Lukas Schwingshackl:** Writing − review & editing, Methodology, Conceptualization. **Jan L. Brozek:** Writing − review & editing, Writing − original draft, Methodology, Conceptualization. **Srinivasa Vittal Katikireddi:** Writing − review & editing. **Zachary Munn:** Writing − review & editing. **Hugh Sharma Waddington:** Writing − review & editing, Conceptualization. **Kevin C. Wilson:** Writing − review & editing, Conceptualization. **Joerg Meerpohl:** Writing − review & editing, Conceptualization. **Daniel Morales:** Writing − review & editing. **Ignacio Neumann:** Writing − review & editing, Methodology, Conceptualization. **Peter Tugwell:** Writing − review & editing, Conceptualization. **Gordon Guyatt:** Writing − review & editing, Methodology, Conceptualization. **Holger J. Schünemann:** Writing − review & editing, Writing − original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

All authors are members of the GRADE working group. There are no competing interests for any author.

Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2025.112086.

Data availability

No data was used for the research described in the article.

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